

## EFY Buying Tips

Checklist for selecting an electric motor and drive system.

- Study relevant Indian Standards.
- Check whether motors and drives are easy to replace and/or repair, and whether technical support is readily available.
- Installation of motors and drives will affect the surrounding environment, depending on factors like heat dissipation and space availability for proper installation.
- Make correct assessment of load, speed, torque requirements and duty cycle.
- Select the enclosure as per the environment where the motor will be used.
- Select an efficient motor to save on power bills.
- Check if protection arrangement is provided along with classification of insulation.
- Manufacturer/vendor credibility is of utmost importance for selection of a reliable motor.
- Use CanMOST, Canadian Motor Selection Tool, to identify the best motor for your needs.
- Variable frequency drives regulate motor speed to match load demand.
- When you select a new energy-efficient motor, be sure that the new motor's power factor is higher than that of the old unit.
- A stepper motor offers the best control over motor speed and load specifically due to its digital nature, but it is more expensive than an AC or a DC motor and needs more expensive drive and motor control systems.



Fig. 5: DC motor drives



Fig. 6: Servo motor and drive

speed is the speed at which the motor runs when fully loaded and is supplied with rated voltage. Common rpm ratings are 3600rpm (two-pole), 1800rpm (four-pole) and 1200rpm (six-pole).

Most motors are directly coupled to the driven equipment when speed of the motor and the driven equipment are the same. In other cases devices like gearbox, chains or belts are introduced between the motor and driven equipment.

**Efficiency.** Motor efficiency requires due attention when selecting a motor for a specific application. Motors running continuously should be as efficient as possible to reduce power consumption. Improvement of even one per cent in efficiency results in savings of enormous power over life span of the motor and thus the cost of power. Standards are defined in IS 12615:2011; these are IE2 - high, IE3 - premium and IE4 - super premium.

### Duty cycle.

Selecting a proper electric motor depends on load pattern, which can be steady or variable over a fixed time duration, following a repetitive cycle

of variation or load with pulsating torque or shocks.

Duty cycle is a fixed repetitive load pattern over a given period of time, and is expressed as the ratio of on-time to cycle period. Frequent starts shorten the life because inrush current at start up heats the conductor rapidly. Duty cycle is also a factor in sizing electric motors.

**Voltage and frequency.** Location where the motor is used dictates voltage and frequency requirements. Supply voltage could be single- or three-phase. It is common to have three-phase motors built in dual voltage ratings of 230/460 volts, 50Hz.

**Enclosures.** A motor's enclosure not only holds its components together, it also protects the internal components from moisture and contaminants. The degree of protection and motor's cooling depends on enclosure type. There are two kinds of enclosures: open and closed.

Open-enclosure motors are designated as drip-proof. An open drip-proof motor allows free exchange of air from outside the motor to circulate around the winding, while being unaffected by drops of liquid or particles that strike or enter the enclosure at any angle from zero to 15 degrees downward from the vertical.

Enclosed category includes three types: totally-enclosed non-ventilated, totally-enclosed fan-cooled and explosion-proof. A good enclosure protects from dust, oil, water and high-pressure liquids so that the motor can be used indoors and outdoors. These are suitable for operation in harsh environments.

**Frame size.** International standards for frame size follow IEC classifications. Relevant Indian standards are IS1231 and IS2223. HP, rpm and enclosure dictate frame sizes. Frame size determines the distance between foot holes nearest to the shaft and opposite drive-end foot holes. This information is required for mounting of the motor.

**Insulation.** Heat is the biggest enemy of motor. Life of the motor greatly depends on the temperature rise of the windings. Main purpose of insulation is to protect the windings in the slots of the stator against heat.

According to IEEE regulation, classification of electric motors' insulation is an indicator of maximum temperature that insulation can tolerate. These are designated by alphabets; for example, for class A insulation maximum temperature is 105°C, and for class S it is 240°C.

We can see insulation class on the motor's plate. Selection of class of insulation is based on the type of load (pump, lift or conveyor belt), working area's ambient temperature and cooling arrangement.

**Altitude.** This indicates the maximum height above sea level at which the motor will remain within its stated parameters. **EFY**